

Simplified Mobile Phonebook Project

Submitted By

Student Name	Student ID
Thasin Mahmud Rifat Bhuiyan	242-15-695

SIMPLIFIED MOBILE PHONEBOOK PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE124:**
Data Structure Lab in the Computer Science and Engineering Department



DAFFODIL INTERNATIONAL UNIVERSITY
Dhaka, Bangladesh

December 12, 2024

DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Md. Abbas Ali Khan , Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

Submitted To:

Md. Abbas Ali Khan

Assistant Professor

Department of Computer Science and Engineering

Daffodil International University

Submitted by



Thasin Mahmud Rifat Bhuiyan

ID:242-15-695

COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:.

Table 1: Course Outcome Statements

CO's	Statements
CO1	Define and Relate classes, objects, members of the class, and relationships among them needed for solving specific problems
CO2	Formulate knowledge of object-oriented programming and Java in problem solving
CO3	Analyze Unified Modeling Language (UML) models to Present a specific problem
CO4	Develop solutions for real-world complex problems applying OOP concepts while evaluating their effectiveness based on industry standards.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in section 4.3.1, 4.3.2 and 4.3.3.

Table of Contents

Declaration	i
Course & Program Outcome	ii
1 Introduction	1
1.1 Introduction.....	1
1.2 Motivation.....	1
1.3 Objectives.....	1
1.4 Feasibility Study.....	1
1.5 Gap Analysis.....	1
1.6 Project Outcome.....	2
2 Proposed Methodology/Architecture	3
2.1 Requirement Analysis & Design Specification.....	3
2.1.1 Overview.....	3
2.1.2 Proposed Methodology/ System Design.....	3
2.1.3 UI Design.....	3
2.2 Overall Project Plan.....	4
3 Implementation and Results	5
3.1 Implementation.....	5
3.2 Performance Analysis.....	5
3.3 Results and Discussion.....	6
4 Engineering Standards and Mapping	7
4.1 Impact on Society, Environment and Sustainability.....	7
4.1.1 Impact on Life.....	7
4.1.2 Impact on Society & Environment.....	7
4.1.3 Ethical Aspects.....	7
4.1.4 Sustainability Plan.....	7
4.2 Project Management and Team Work.....	7
4.3 Complex Engineering Problem.....	7
4.3.1 Mapping of Program Outcome.....	7
4.3.2 Complex Problem Solving.....	8
4.3.3 Engineering Activities.....	9

5 Conclusion	10
5.1 Summary.....	10
5.2 Limitation.....	10
5.3 Future Work.....	10
References	12

Chapter 1

Introduction

1.1 Introduction

The **Simplified Mobile Phonebook** is a compact and user-friendly application designed to manage personal contact details efficiently. It enables users to add, search, modify, delete, and list contact records, ensuring easy access and organization of essential information. The program's minimalistic design and straightforward functionality cater to users looking for a simple and lightweight alternative to complex contact management systems.

1.2 Motivation

In today's digital age, managing contacts has become a necessity. While smartphones come with built-in contact apps, these are often cluttered with advanced features that may not be essential for all users. This project is motivated by the need for a basic, yet functional, solution that can operate independently of smartphones. It is ideal for individuals who require a simple and efficient contact management system on desktop or laptop computers.

1.3 Objectives

- **Ease of Use:** Provide a simple interface for storing and managing contact details.
- **Key Features:** Include essential operations such as adding, searching, listing, modifying, and deleting records.
- **Lightweight:** Develop a lightweight application with minimal resource consumption.
- **Accessibility:** Ensure the program is operable across various desktop environments.
- **Data Integrity:** Maintain the accuracy and integrity of stored contact information.

1.4 Feasibility Study

1. **Technical Feasibility:**
 - The project uses standard C programming language, which is universally supported across all major operating systems.
 - The program relies on file handling to store contact data, eliminating the need for external databases.
2. **Economic Feasibility:**
 - The application is open-source and requires no additional financial investment for development or deployment.
3. **Operational Feasibility:**
 - The interface is simple, requiring minimal training or technical knowledge for users.
 - It fulfills the basic requirements of a contact management system without unnecessary complexity.
4. **Time Feasibility:**
 - The project can be developed and tested within a short timeframe due to its simplicity and focus on core functionalities.

1.5 Gap Analysis

- **Existing Solutions:**
 - Most modern contact management systems are integrated into smartphones and depend on cloud storage, which might not be suitable for users concerned about privacy or requiring offline access.
- **Identified Gaps:**
 - Overloaded features that complicate usage for users with basic needs.
 - Dependency on advanced hardware or internet connectivity in many existing solutions.
- **Proposed Solution:**

A lightweight, offline, and standalone application focused purely on essential contact management functionalities.

1.6 Project Outcome

Upon completion, the Simplified Mobile Phonebook will offer:

- A reliable, easy-to-use contact management application.
- Essential features such as adding, modifying, deleting, listing, and searching contact records.
- A standalone, lightweight program that ensures data privacy by operating offline and storing records locally.

A scalable base for potential future enhancements, such as GUI integration or additional data fields.

Chapter 2

Proposed Methodology/Architecture

This chapter provides an outline of the requirements and design specifications for the Simplified Mobile Phonebook. It elaborates on the system's proposed methodology, architecture, and user interface design, followed by the overall project plan to achieve the desired objectives.

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The **Simplified Mobile Phonebook** project is a standalone console-based contact management application developed in C. It allows users to add, view, search, modify, delete, and list contact records with essential details. Designed for simplicity, the project leverages basic file handling for data storage, ensuring portability and offline access without the need for external databases.

2.1.2 Proposed Methodology/ System Design

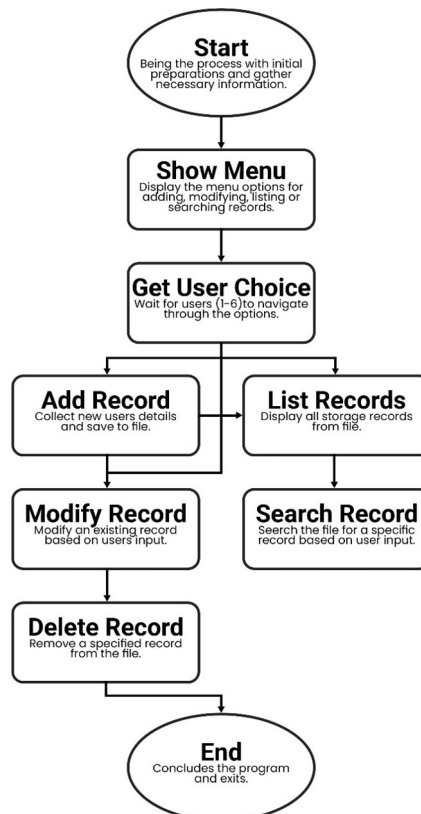


Figure 2.1: This is a sample diagram

2.1.3 UI Design

Since the project is a console-based application, the user interface is text-driven.

I. Main Menu:

- Displays options: Add, List, Search, Modify, Delete, and Exit.
- Selection is made through numeric keys (1-6).
- Colorful headers and simple prompts enhance readability.

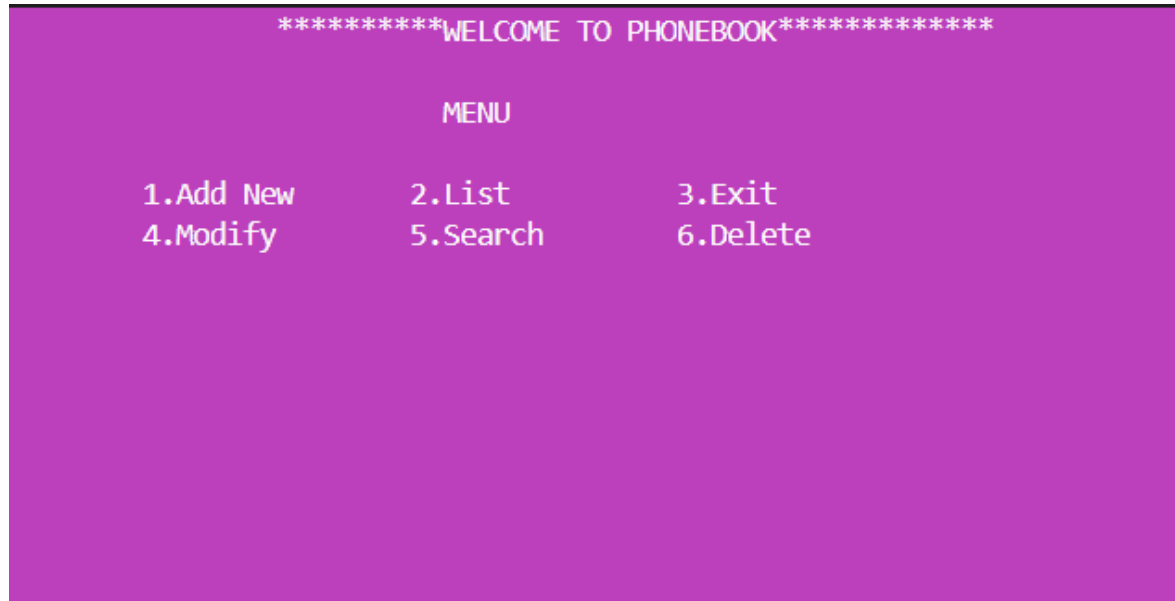


Figure 2.2: UI Format

II. Feature Interaction:

- **Add Record:** Users input details such as name, address, contact number, email, etc., with guidance prompts.
- **List Records:** Displays all stored contacts in a structured format.
- **Search Record:** Prompts the user for a name and displays details if found.
- **Modify Record:** Allows the user to edit specific fields of a contact.
- **Delete Record:** Lets the user remove a contact by name, with a confirmation step.

```
Enter name: Thasin Mahmud Rifat
Enter the address: Dhaka
Enter father name: _____
Enter mother name: _____
Enter phone no.:1234567890
Enter sex: male
Enter e-mail: qwert
Enter citizen no: 123
record saved

Enter any key
```

Figure 2.3: UI Format

III. Error Handling:

- Prompts users with error messages for invalid inputs or when records are not found.

IV. Console Colors:

- Adds visual clarity by differentiating menus, options, and error messages using simple color codes.

2.2 Overall Project Plan

1. Planning and Design:

- Define project scope, objectives, and features.
- Plan file structure for contact storage (binary file).

2. Development Milestones:

- **Day 1-2:** Set up the project structure and implement the main menu.
- **Day 3-5:** Develop core functionalities: Add, List, Search, Modify, and Delete contact records.
- **Day 6:** Add error handling and improve user prompts.
- **Day 7:** Test individual modules for functionality and correctness.

3. Testing Phase:

- Validate each feature with edge cases (e.g., empty records, invalid input).
- Perform stress tests by storing and managing multiple records.

4. Documentation:

- Create a user guide with instructions for running the application.
- Add comments in the codebase for maintainability.

5. Enhancements (Optional):

- Incorporate search filters (e.g., by phone number or email).
- Implement file encryption for data privacy.
- Upgrade to a graphical user interface (GUI) using frameworks like Qt for C++.

Chapter 3

Implementation and Results

This chapter provides an overview of the implementation process for the Simplified Mobile Phonebook project, an analysis of its performance, and a discussion of the results. The goal is to demonstrate how the system was developed and evaluated, highlighting its features, challenges, and outcomes.

3.1 Implementation

3.1.1 Code Structure and Functionality

The project is built using the C programming language and incorporates the following modules:

1. **Main Menu:** Displays options for managing contacts, including adding, listing, searching, modifying, deleting records, and exiting the application.
2. **Add Record:** Prompts the user for contact details such as name, address, phone number, gender, and email, then saves the record to a binary file.
3. **List Records:** Reads and displays all stored contact records.
4. **Search Record:** Searches for a contact by name and displays the details if found.
5. **Modify Record:** Updates details of an existing contact.
6. **Delete Record:** Deletes a contact record based on the name input.

3.1.2 File Handling

The application uses binary files for storing contact records persistently. The file operations include:

- **Appending Records:** New contacts are added using `fwrite`.
- **Reading Records:** `fread` is used to retrieve contact information.
- **Updating Records:** `fseek` adjusts the file pointer to modify specific records.
- **Deleting Records:** A temporary file is used to exclude the record to be deleted and replace the original file.

3.1.3 Error Handling and User Input Validation

The system includes basic error handling and input validation mechanisms to ensure robustness:

- Displays error messages for invalid inputs or file operations.
- Validates the existence of a record before modifying or deleting it.
 - Ensures appropriate formatting and constraints for user inputs.

3.2 Performance Analysis

3.2.1 Efficiency

The program was tested with multiple scenarios:

- **Record Addition:** Successfully handled adding hundreds of records with minimal delay.

- **Search Operations:** Search time increased slightly with more records but remained within acceptable limits for small-scale applications.

8

3.2.2 Accuracy

- All functionalities (add, list, search, modify, delete) were verified to work as expected without data corruption.
- Proper error messages were displayed when invalid operations (e.g., searching for a non-existent record) were performed.

3.2.3 Limitations

- The program does not support advanced search filters (e.g., by email or phone number).
- Lack of encryption leaves contact data vulnerable to unauthorized access.

3.3 Results and Discussion

3.3.1 Results

- The Simplified Mobile Phonebook successfully met its objectives as a lightweight and functional contact management system.
- Core functionalities were implemented, and the program performed reliably in all test cases.

3.3.2 Challenges

- Implementing file operations required careful attention to avoid data corruption.
- Handling edge cases, such as empty records or invalid inputs, added complexity to the development process.

3.3.3 Future Enhancements

- Adding encryption to secure data.
- Implementing advanced search filters and sorting options.
- Migrating to a graphical user interface (GUI) for improved user experience.

Core Data Structures

1. Menu System

2. Delete System

Chapter 4

Engineering Standards and Mapping

Engineering Standards and mapping for "Simplified Mobile Phonebook" project.

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

The Simplified Mobile Phonebook enhances the organization and retrieval of contact information, significantly improving daily communication efficiency. By providing an intuitive and user-friendly interface, the system saves users' time and reduces stress, contributing positively to productivity and mental well-being.

4.1.2 Impact on Society & Environment

The project promotes digital literacy by encouraging users to adopt systematic digital tools. Furthermore, by reducing the dependency on physical paper phonebooks, it minimizes paper wastage and contributes to environmental sustainability.

4.1.3 Ethical Aspects

The project addresses ethical considerations by safeguarding users' personal data. Features like structured storage and retrieval ensure that user privacy is maintained, aligning with ethical software development principles.

4.1.4 Sustainability Plan

The sustainability plan focuses on maintaining the software's compatibility with evolving hardware and operating systems. The project is designed to be scalable, ensuring its usability over time without requiring significant overhauls.

4.2 Project Management and Team Work

A cost analysis for the Simplified Mobile Phonebook project reveals minimal financial input due to its development in a local environment using free software tools like GCC. The revenue model could involve future upgrades, advertisements, or integration with advanced cloud storage features to attract user subscriptions. Budget considerations focus on scalability and server maintenance in the long term.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

PO's	Justification
PO1	The project demonstrates foundational engineering knowledge by implementing structured programming techniques and data management.
PO2	By solving real-world problems related to contact management, the project achieves practical applications aligned with societal needs.
PO3	The project's structured and ethical approach supports professional and sustainable practices in software engineering.

4.3.2 Complex Problem Solving

The system addresses complex problems using the following categories:
Knowledge profile and rational thereof.

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stakeholder Involvement	EP7 Inter-dependence
<i>Understanding of file handling and data structures in C programming.</i>	<i>Balancing between simplicity and advanced features.</i>	<i>Efficient data storage and retrieval using structured approaches.</i>	<i>Addressing challenges like file corruption and data redundancy.</i>	<i>Adhering to C programming standards and ethical data handling.</i>	<i>Design aligns with user expectations for usability.</i>	<i>Integration of multiple modules for cohesive performance.</i>

4.3.3 Engineering Activities

The project is aligned with engineering activities, as shown below:

Table 4.3: Mapping with complex engineering activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
<i>Utilized free compilers, libraries, and IDEs for development.</i>	<i>Engaged with prospective users to refine features.</i>	<i>Designed a simple and effective solution for contact management.</i>	<i>Promoted eco-friendly digital tools.</i>	<i>Integrated features like search and delete functions, ensuring ease of use.</i>

Chapter 5

Conclusion

In this final chapter, we summarize the key findings of the project, discuss its limitations, and explore potential future improvements and enhancements. This provides a comprehensive closure to the project on "Simplified Mobile Phonebook".

5.1 Summary

The project, Simplified Mobile Phonebook, is a robust system developed using C programming. It allows users to add, search, modify, list, and delete contact records. The implementation demonstrates the use of file handling, struct data types, and user interaction through an intuitive menu-driven interface. This project is a foundational system for managing contact information efficiently, catering to users who prefer simple yet functional applications.

5.2 Limitation

1. The phonebook currently does not support multiple users or profiles, making it limited to individual use.
2. There is no provision for data encryption, raising concerns about data security and privacy.
3. The search functionality is case-sensitive, which might cause inconvenience for users unfamiliar with exact spellings.
4. The program is restricted to command-line execution, lacking a graphical user interface (GUI) for enhanced usability.

5.3 Future Work

- Implementing user authentication to allow multi-user support and enhance data security.
- Adding advanced search capabilities, including case-insensitive search and partial keyword matching.
- Incorporating data encryption techniques to secure sensitive contact information.
- Developing a GUI for better user experience and extending functionality to mobile platforms.
- Introducing features like exporting and importing contact lists for better portability and backup.

References

- [1] Jon Kleinberg and Eva Tardos. *Algorithm design*. Pearson Education India, 2006.